→javascript

→html is a markup language

→css is a styling language

→javascript is a programming language

Using the console:

Uses repl

Repl:read-evaluate-print-loop

Read→reads input

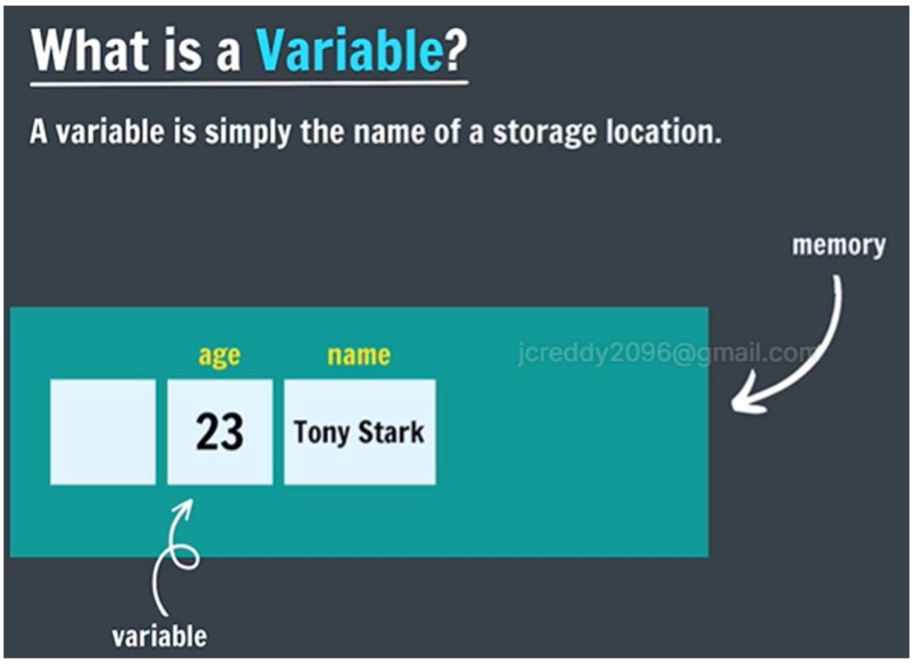
Evaluate→calculation

Print→it prints

Loop→repeat

→by clicking ctrl+L it will clear the warnings in the console

→what is a variable?



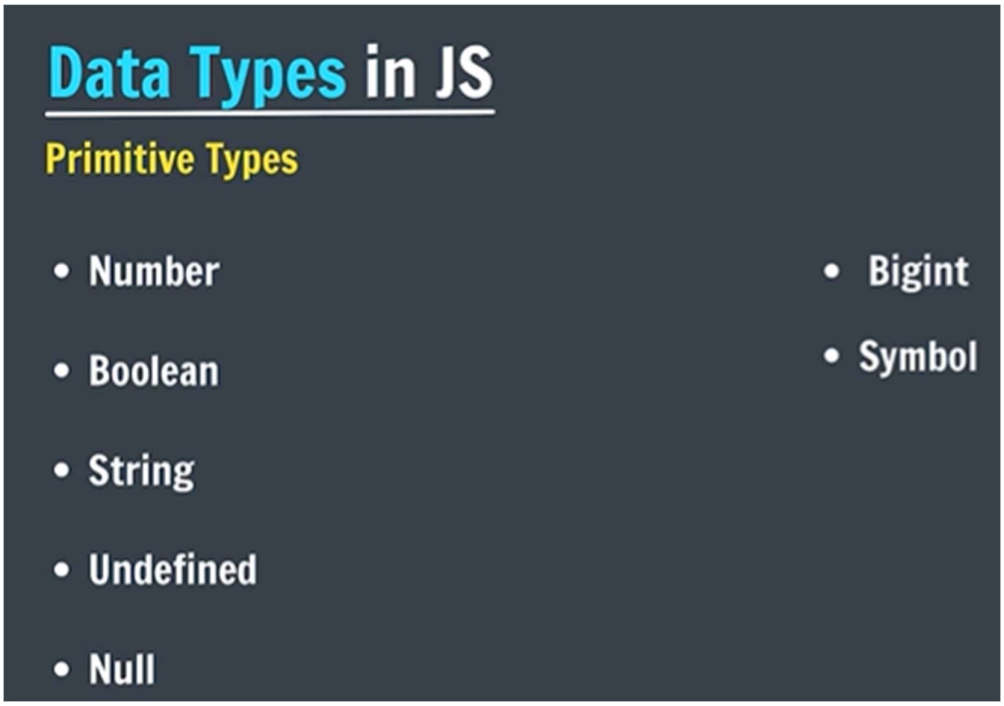
→variable means we change the value of the variable

Ex: a=10,b=20

a=30;

name= “tony stark”

→data types in js



In console if we type typeof(a) a is a variable by this we can know the data type of the variable.

→numbers in js

- Positive (14) & Negative (-4)
- Integers (45, -50)
- Floating numbers - with decimal (4.6, -8.9)

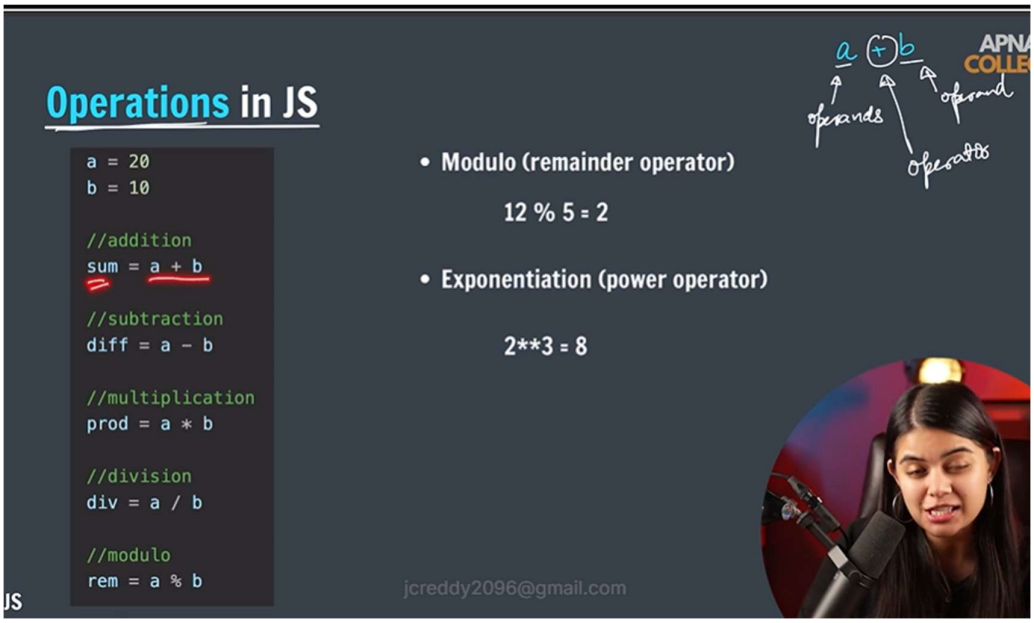
We can do the operations in js

```
> a=10
< 10
> b=5
< 5
> a+b
< 15
> a-b
< 5
> a*b
< 50
> a/b
< 2
> a%b
< 0
```

a+b,a-b,a/b,a*b

```
> 0.999999999999
< 0.999999999999
> 0.99999999999999999999999999999999999999
< 1
```

→operations in js



a+b

a,b are operands

+ represents the operation

→to get the remainder of any number we find the modulo(%) of that number

→by using the modulo operator we use to find whether the number is even or odd number

→for an even number we get final =0;

→for an odd number we get =1;

Ex: 4%2=0, 3%2=1

→Exponentiation(power operator)

a**b

a=2,b=3

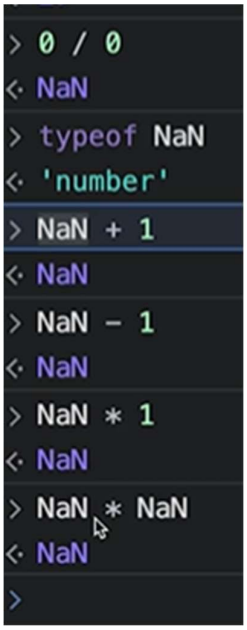
2**3=8

→NaN in js

NaN is Not A Number

The NaN global property is a value representing Not-A-Number.

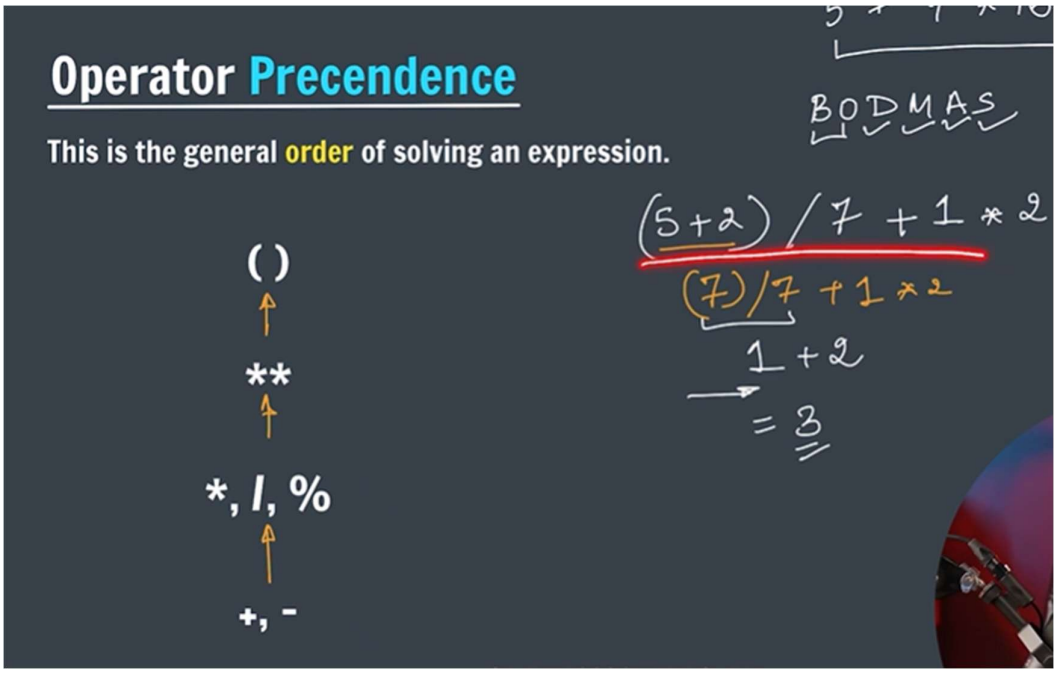
```
0 / 0

NaN - 1

NaN * 1

NaN + NaN
```

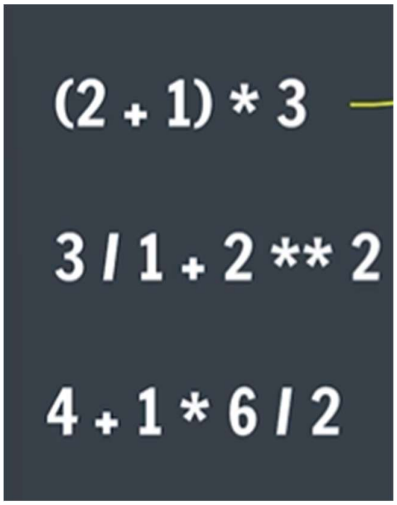


→Operator precedence

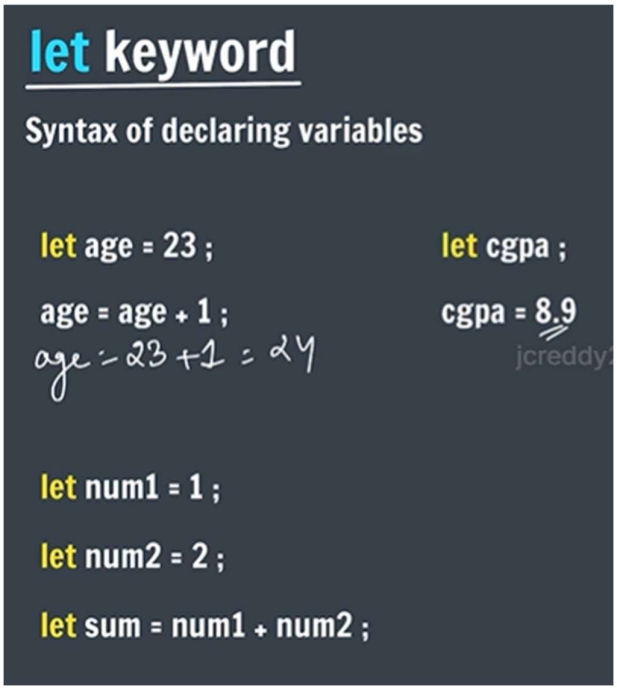
→in operator precedence we take the precedence based on the level top to bottom and then if they are at same level like *,/,% at that time we need to move left to right to solve.

Left to right

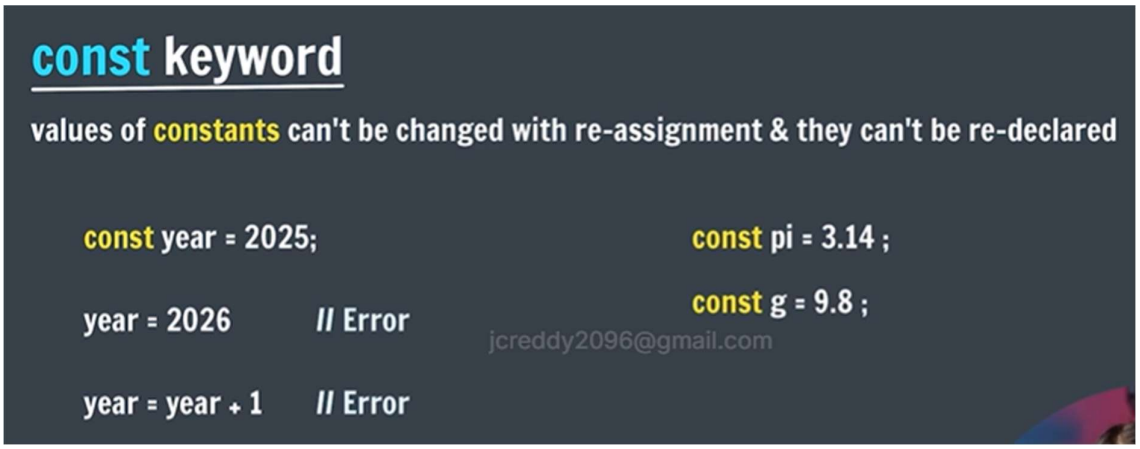




→let keyword



→const keyword



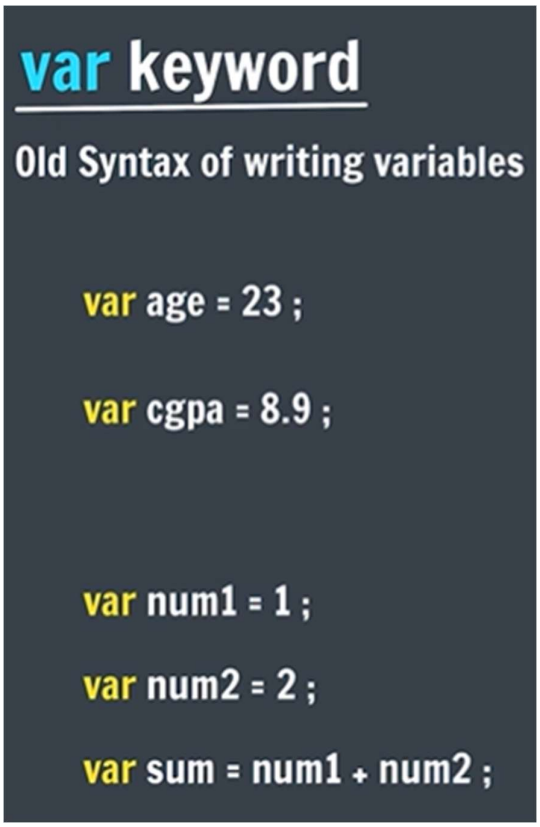
→once we declare the variable under const it value cannot be changed throught the program .

→once its value is fixed that's it.

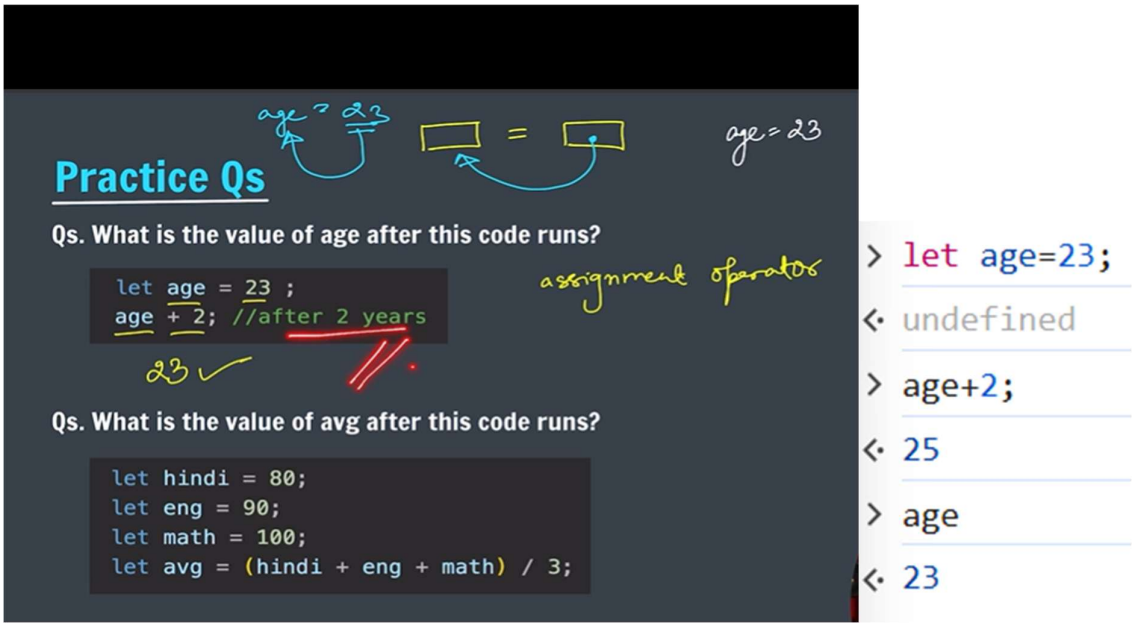
→in let we can change the values at any line of the program by using the let for any variable.

```
> const pi = 3.14;
< undefined
> let rad = 4;
< undefined
> let area = pi * r ** 2;
✕ ▶ Uncaught ReferenceError:
      at <anonymous>:1:17
> let area = pi * rad ** 2;
< undefined
> area
< 50.24
```

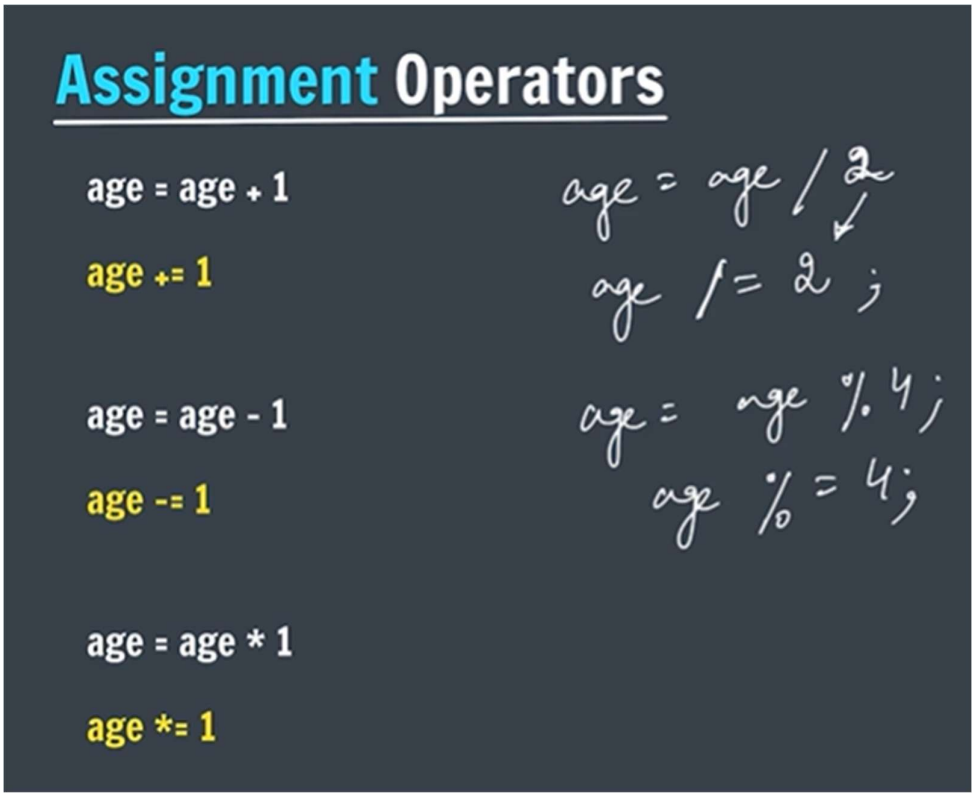
→var keyword: var keyword works as it is like the let keyword but the var is outdated usage.



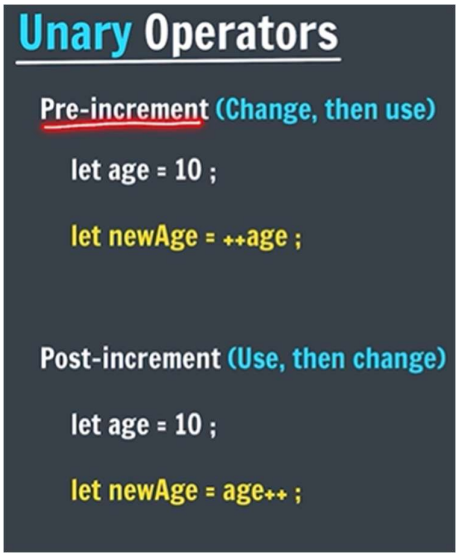
→practice question

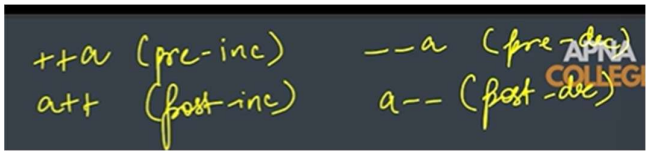


→assignment operators

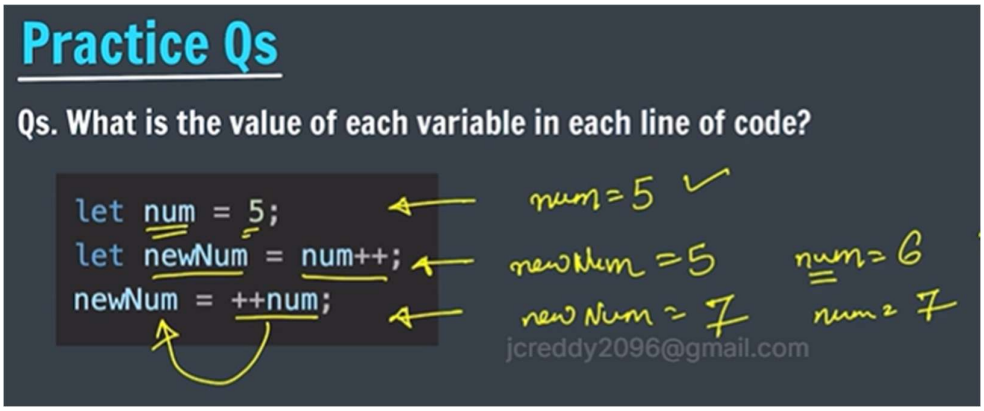


→unary operators

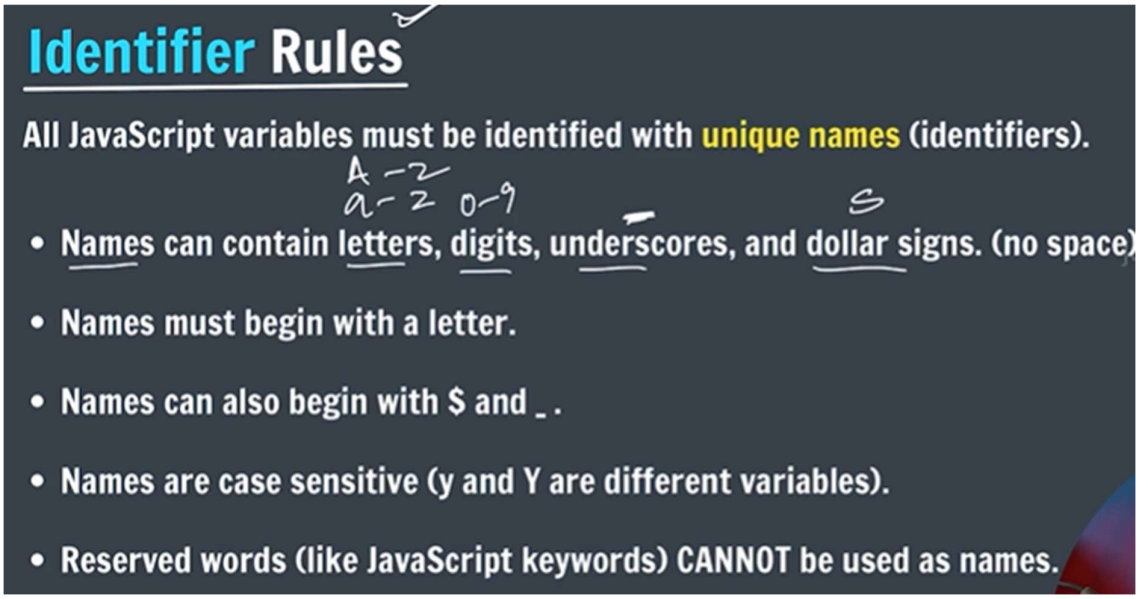


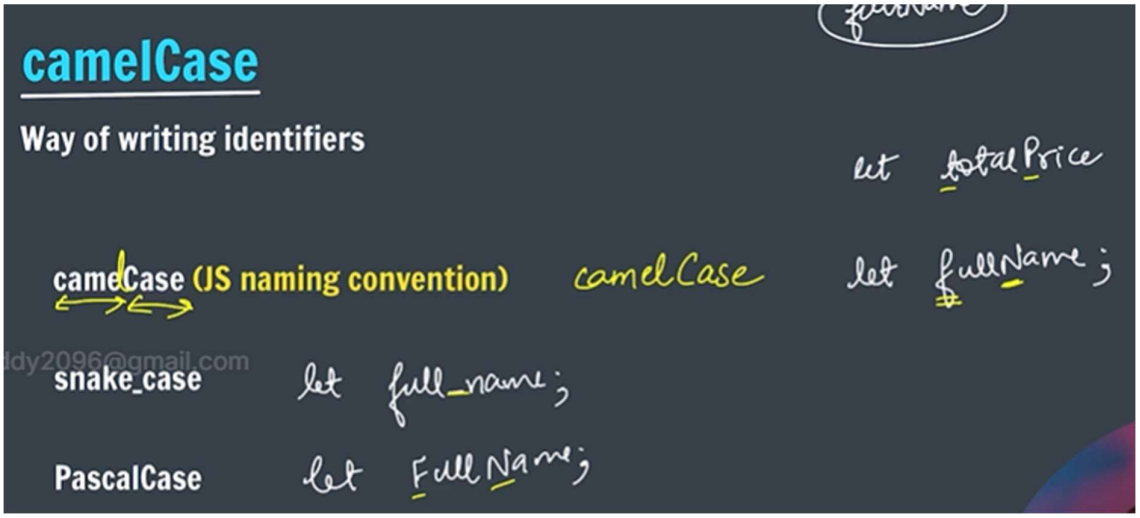


→practice qs

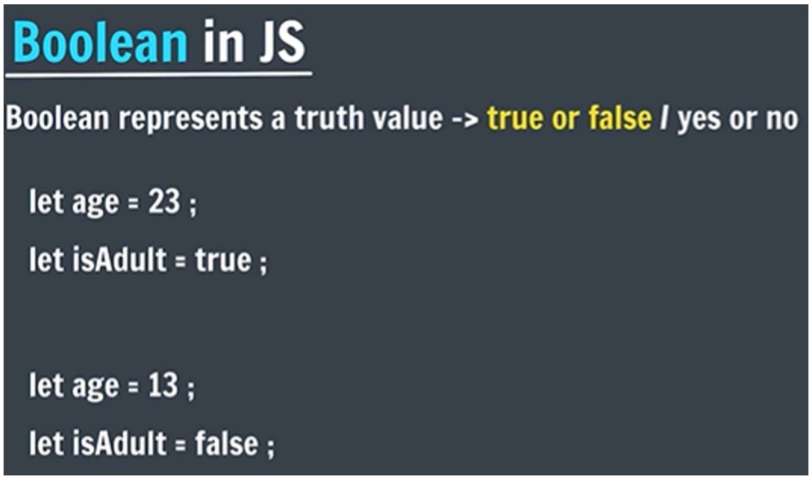


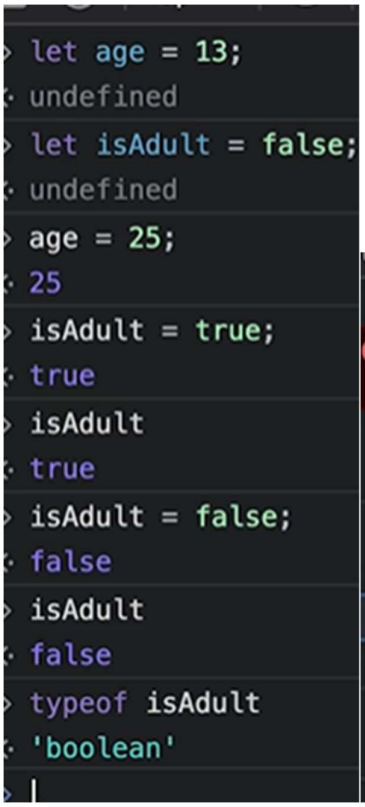
→identifier rules

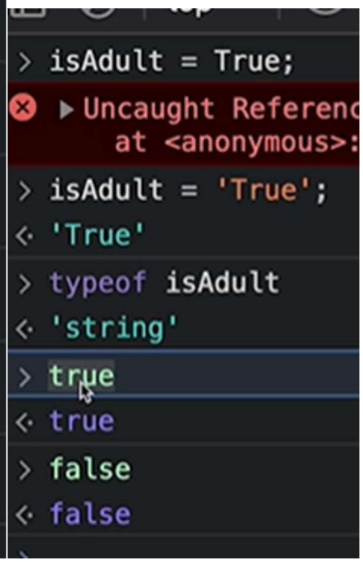




→boolean in js

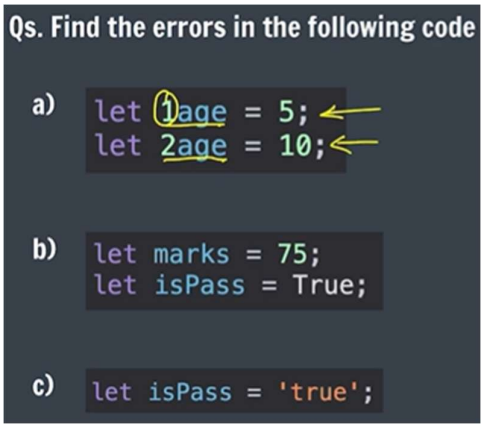


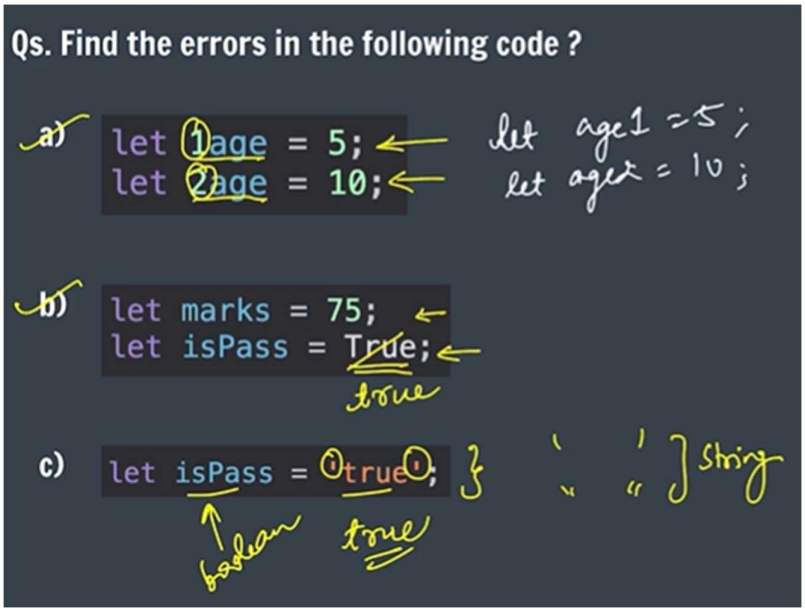




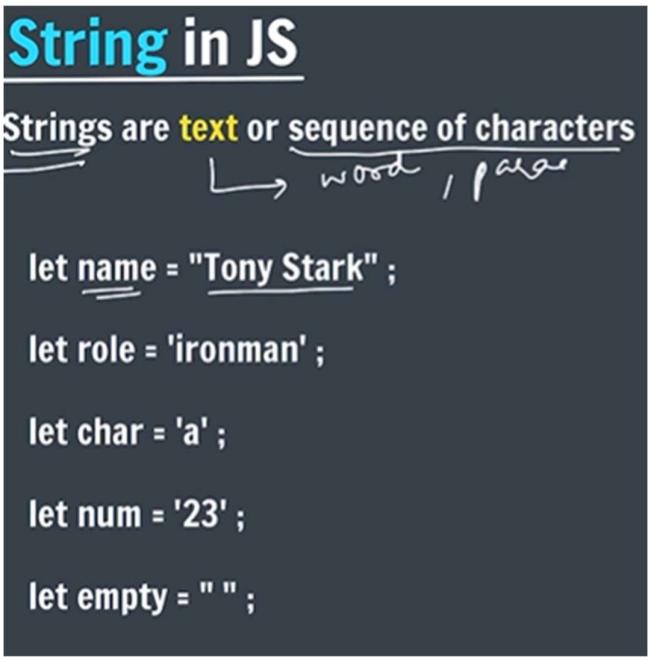
→in js we can also change the type of a variable by changing its value in js unlike c++,java from program starting to end same type variable we use but in js we can change.

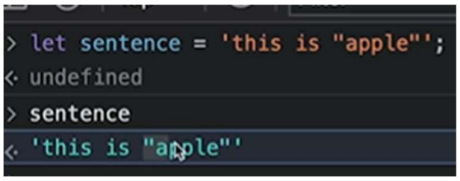
→practice questions





→String in js

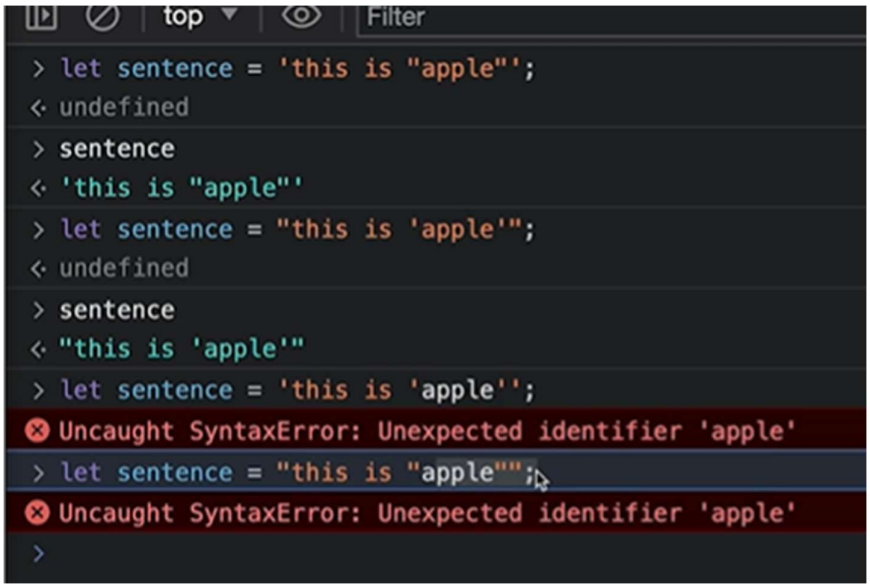




→in strings we can use the single quotes(‘’) and double quotes (“”) for writing the strings.

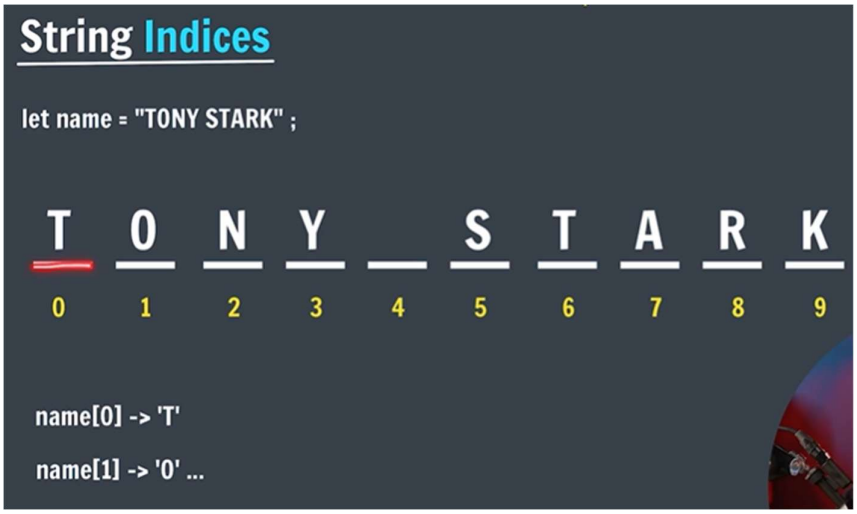
→let a=”” →it is also a string

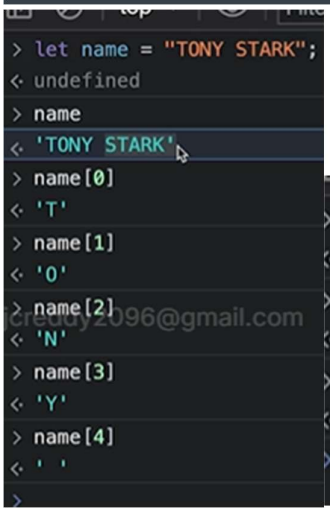
Let b=” “→it is also a string

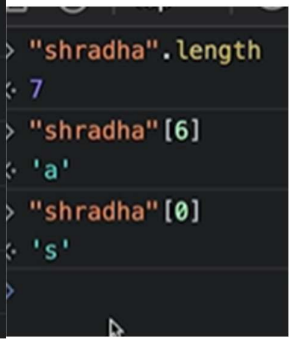


→string indices

→index=position



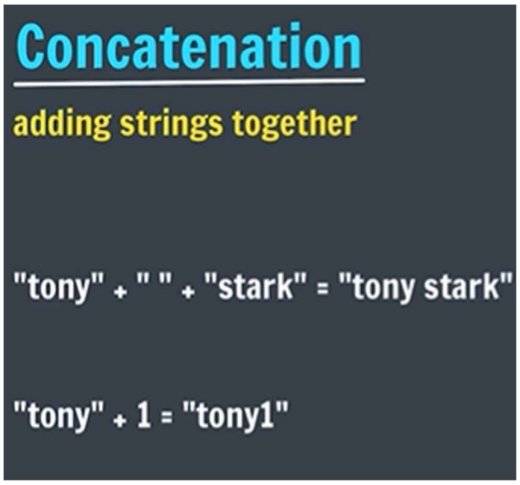




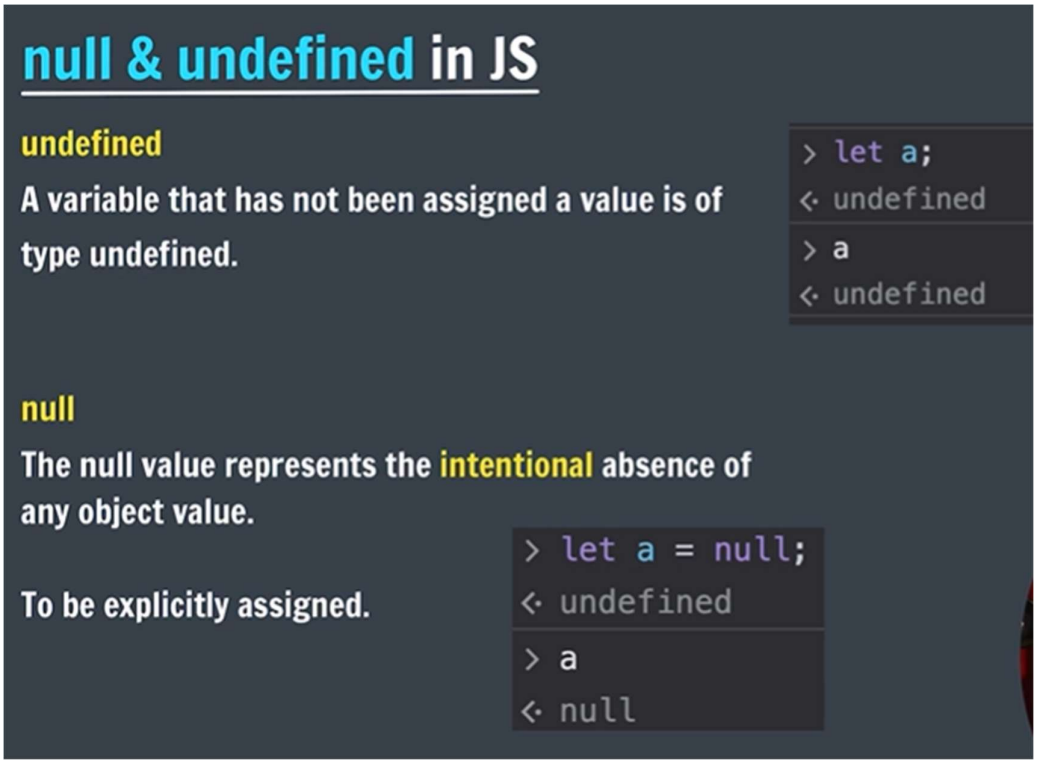
→String length

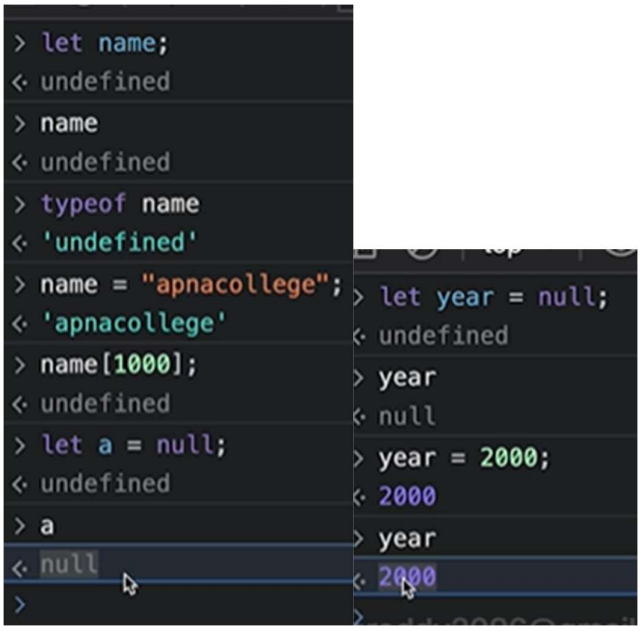
→to know the length of the variable name.length

→string concatenation



→null & undefined in js



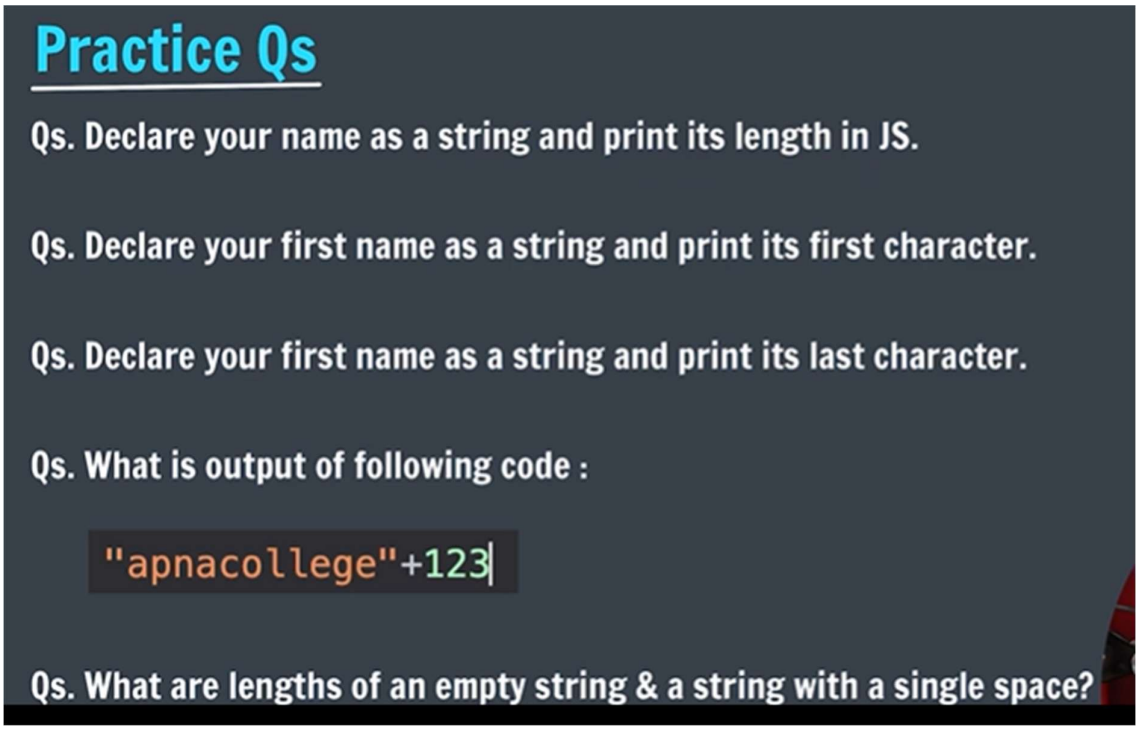


→undefined : means js doesn't know the value of that variable

→null: this value we assigned intentionally in program we assigh another value to it

→null is a keyword.

→practice questions

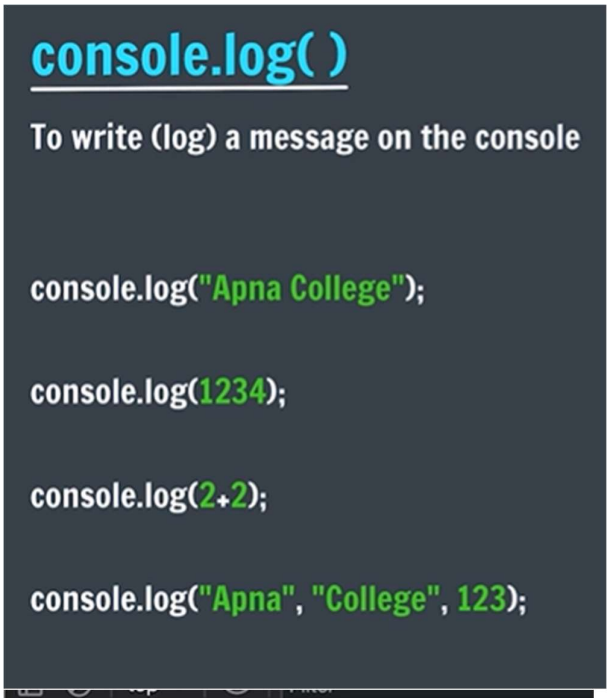


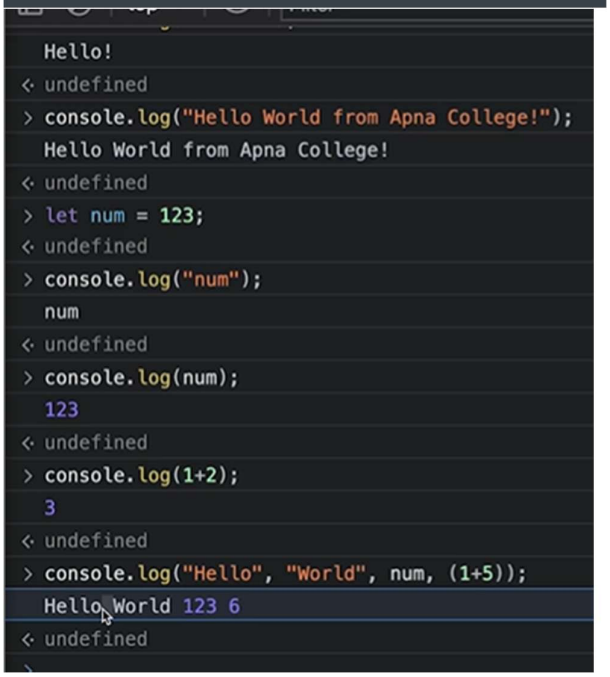
```
let name="subbu"
undefined
name
'subbu'
name.length
5
let firstname="jayachandra"
undefined
firstname
'jayachandra'
firstname[firstname.length-1]
'a'
firstname[0];
'j'
let n=""
undefined
n.length
0
let a=" ";
undefined
a.length
1
```

```
> let "apnacollege"+123;
Uncaught SyntaxError: Unexpected string
> let "apnacollege"+123
Uncaught SyntaxError: Unexpected string
> let n="apnacollege"+123
undefined
> n
'apnacollege123'
>
```

→js-2

Console.log(“hello”);





→js file doesn’t directly connected to browser by using the console.log we use to print the values in browser using the console.log()

→linking js file

The correct position to add the script tag in the html file is not in the head part it is in the body tag at the ending of the html code we add the script tag.